LESSON 9 • TERM 1 • LEVEL 1

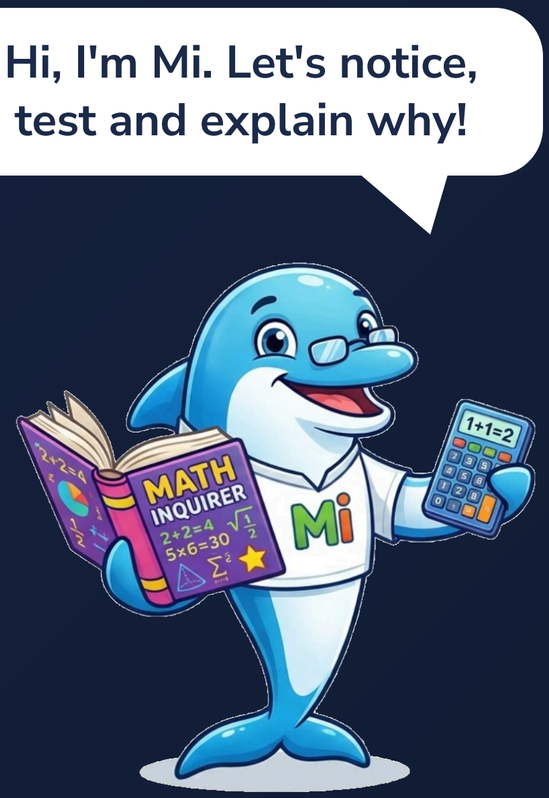
# Draw and Measure

Length: centimetres

TODAY WE WILL

Use a ruler to measure and to draw lines exactly.



4–10 MIN

# Warm-up: measure with clips

1. Lay clips along your pencil, end to end.
2. No gaps. No overlaps.
3. Count the clips.

**WHAT IF?**

Your friend uses bigger clips. Will the number be the same?

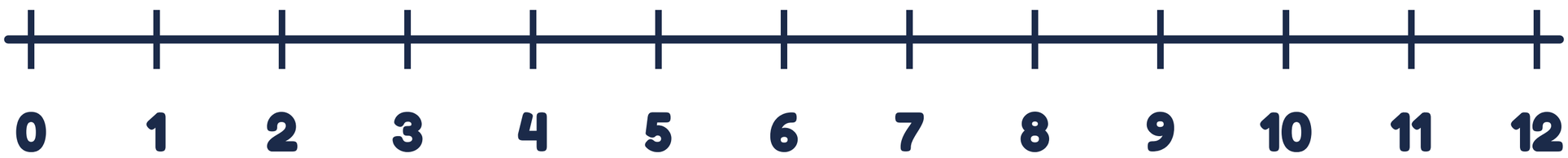



Rulers ready? Off we go!

4–10 MIN

# Find 0 on your ruler

Put your finger on the 0 mark. Count along the marks.

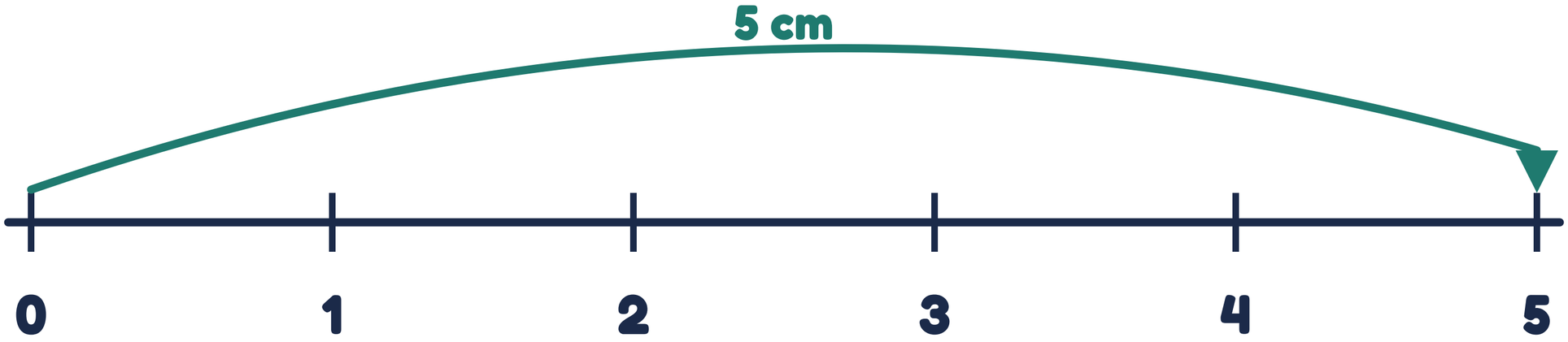



Say the numbers with me

4–10 MIN

# One centimetre

The gap between two marks is 1 cm.




Slide your finger from 0 to 5. How many gaps? How many cm?

# Sneak-peek puzzles

10–14 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Draw a dot.

Use your ruler.

Draw a line from the dot, exactly 3 cm long.

PUZZLE 2: think

Ali puts one end of a pencil at the 1 mark of his ruler.

The other end is at the 9 mark.

Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.

# Explore: guess, then measure

14–28 MIN

1. Find 3 things: a pencil, a spoon, a rubber or a toy.
2. Guess how many cm long each one is.
3. Put one end at the 0 mark. Read the mark at the other end.
4. Say: my ___ is about ___ cm.

**WHAT IF?**

The end is between two marks. Which mark is closer?

14–28 MIN

# Draw it!

1. Put a dot where the line starts.
2. Put the 0 mark on the dot.
3. Make a small mark at the end.
4. Join them with a straight line.

**3 cm** | **8 cm** | **12 cm**

14–28 MIN

# What did we notice?

- Where do we start measuring?
- What do we do when the end is between two marks?
- Why do we all use centimetres?

# The parts of a ruler

28–38 MIN




REMEMBER

0 is where we start.

The gap from one mark to the next is 1 cm.

28–38 MIN

# Measure: three steps

1. Put one end of the thing at the 0 mark.
2. Keep the ruler straight along the thing.
3. Read the mark at the other end. That is the length in cm.

**Look from straight above**

28–38 MIN

# Draw a line 7 cm long

1. Put a dot where the line starts.
2. Put the 0 mark on the dot.
3. Make a small mark at the 7.
4. Join them with a straight line.
5. Check: 0 at the start, 7 at the end.

28–38 MIN

# Between two marks

The end is between the 8 mark and the 9 mark. It is closer to the 9 mark.

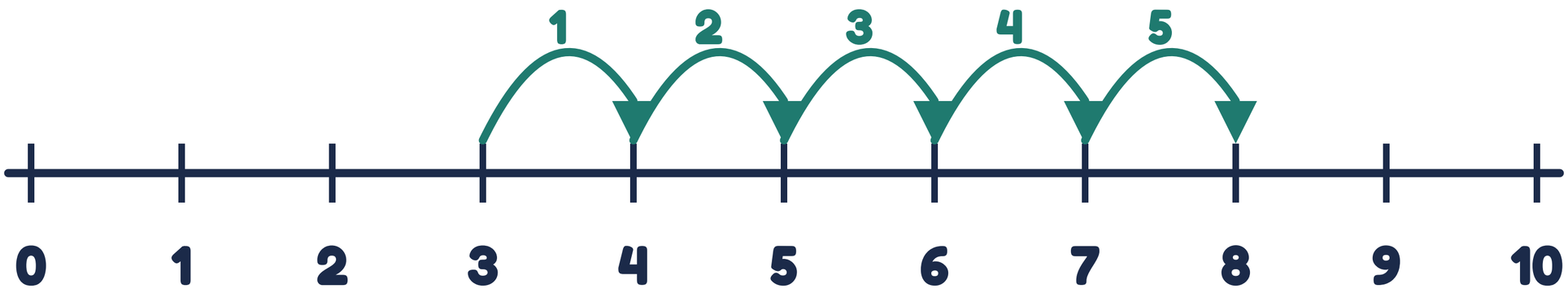



NEAREST CENTIMETRE

Choose the closer mark. This one is about 9 cm.

28–38 MIN

# A line that starts at 3




The line starts at 3 and ends at 8. Count the jumps. 5 jumps. The line is 5 cm long.

**8 take away 3 is 5**

28–38 MIN

# Your turn

**Write on your paper. About 1 minute**

DRAW

Draw a line 5 cm long.

WORK OUT

A line goes from the 2 mark to the 9 mark.

How long is it?

28–38 MIN

# Check: your turn

DRAW

The line starts at the 0 mark and ends at the 5 mark.

WORK OUT

9 take away 2 is 7.

**7 cm**

38–42 MIN

# Jump the tens, hop the ones

Pretend a big jump is 10 cm. A small hop is 1 cm.

14 23 30 8

14 cm is 1 big jump and 4 small hops. Sip some water.

42–70 MIN

# Workbook page 1: Start here

Questions 1 to 3. About 9 minutes. Use your ruler

**S1** In the box, draw a line 7 cm long. Start at the 0 mark.

**S2** In the box, draw a line 9 cm long. Ali's line is 4 cm long.

**S3** Measure the length of your pencil. Write your guess first. Then measure it in whole cm.

42–70 MIN

# Check page 1

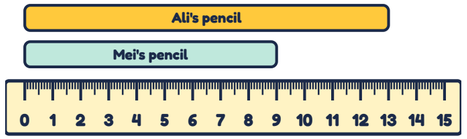
| | |
|---|---|
| S1 | The line ends at the 7 mark (it starts at the 0 mark and is 7 cm long) |
| S2 | 5 cm longer (9 take away 4) |
| S3 | OPEN ANSWER, accept any sensible measure (guess and measure within about 3 cm of each other); a new pencil is about 17 to 19 cm. A grown-up checks with the pencil: the 0 mark at one end, read the mark at the other end |

# Workbook page 2: Start here

42–70 MIN

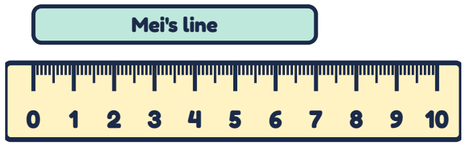
Questions 4 to 6. About 8 minutes. Use your ruler

**S4** Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.




**S5** Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.

**S6** Look at Mei's line on the ruler. Mei says it is 8 cm long.



# Check page 2

42–70 MIN

| | |
|---|---|
| S4 | 13 cm; 9 cm; Ali's is 4 cm longer |
| S5 | The second line is 8 cm long (5 + 3) |
| S6 | No. Her line is 7 cm. She should draw on to the 8 mark |

42–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Our idea in a new setting |
| **Go further (page 4)** | Bigger puzzles |
| **Investigation (page 5)** | Does the start mark matter? |
| **Check** | Finish and check your own work |

**Everyone may choose. You can swap at any time.**

# Talk and explain

70–76 MIN

MEI AND BEN MEASURE THE SAME PENCIL

Mei says 15 cm.

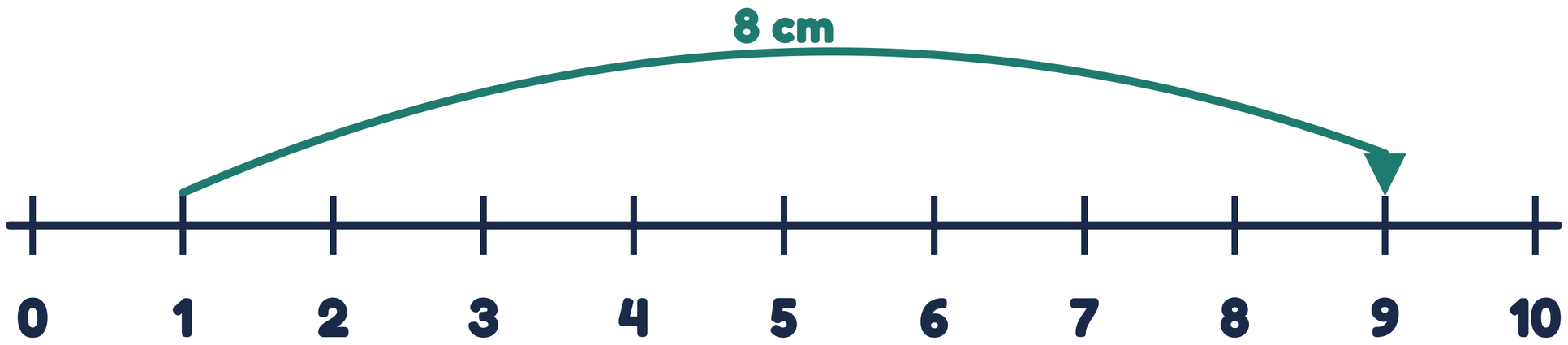
Ben says 16 cm.

Why might their answers be different? How can we check?

I think … because …

70–76 MIN

# Back to the sneak-peek




**No! The pencil starts at the 1 mark, so it is 8 cm long: 9 take away 1.**

# Exit ticket

76–82 MIN

On your own. No looking back. About 3 minutes

**E1** In the box, draw a line 6 cm long. Start at the 0 mark.

**E2** Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

**E3** A line starts at the 3 mark and ends at the 10 mark. How long is the line?

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | A line from the 0 mark to the 6 mark |
| E2 | 9 cm longer |
| E3 | 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10) |

# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ starting at the 0 mark ☐ drawing a line exactly<br>☐ a line that starts at another mark ☐ adding the pieces of a path ☐ something else |
| **What helped me** | ☐ my ruler ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question |
| | ☐ I am still stuck (my tutor will come and help) ☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Story Steps: Read It Twice

You need a pencil and paper.



OPTIONAL

# Go further: how many cuts?

A strip of paper is 10 cm long. You may cut it at the 1 mark, the 2 mark and so on, but not at the ends.

How many places can you cut? What if the strip is 20 cm long?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | A line ends at the 9 mark. Is it always 9 cm long? |
| **Make a puzzle** | Draw a path with 3 straight pieces for your grown-up to measure |
| **Teach it back** | Show how to draw a line 9 cm long, in steps |

**Optional. Everyone may join.**